

Why do Students Avoid Each Other?

Investigating Positive and Negative Network Effects

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Why study avoidance?

- Negative ties can greatly influence positive network dynamics and beliefs and behavior (e.g., Labianca and Brass, 2006; Candollone et al., 2025)
- Many different types of negative ties: dislike, bullying, aggression, and avoidance
 - Broad literature on these (e.g., Huitsing et al., 2014, among others)
 - Less studied
- Avoidance is particularly interesting since it is often more frequent and less socially complex than bullying and aggression (Wójcik and Flak, 2019; Kros et al., 2021)

Our contribution

We use 193 classroom networks to study:

- (1) Structural dependencies in classroom avoidance networks
- (2) How a student's tendency to avoid others and be avoided relates to their position in the classroom friendship network

We extend previous work in three ways:

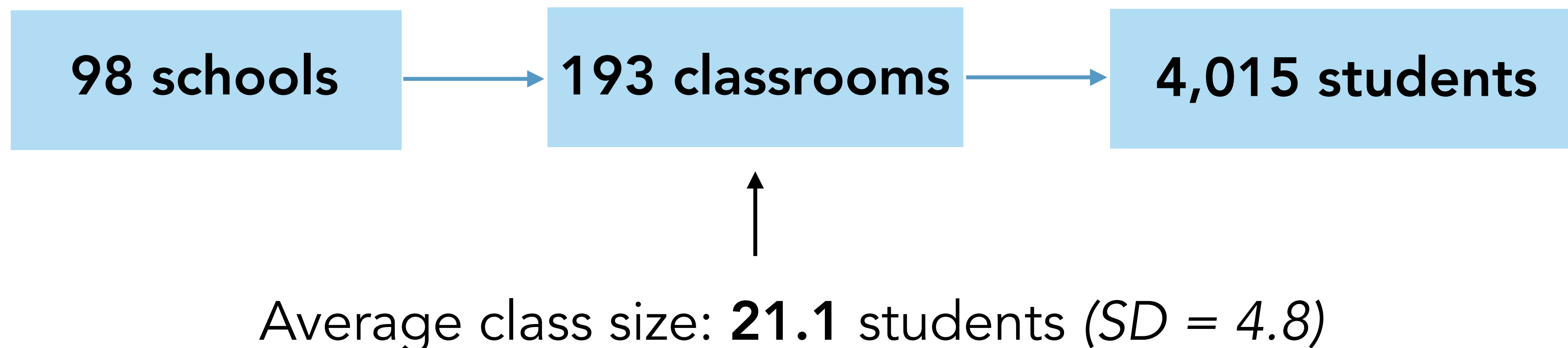
- (1) Account for global network properties in friendship networks (e.g., student connectedness)
- (2) Control for lower-level, non-triadic structures between networks (e.g., tie overlap)
- (3) Use a larger sample of classroom networks

Participants and procedures

Netherlands Wave 1: Children of Immigrants Longitudinal Survey in Four European Countries (CILS4EU)

- Primarily 14-year-old students (2010-2011)
- Questions on students' background and self-reported relationships with classmates

Removing classes with low survey participation (<50%) and small class sizes (<10 students):



Network measures

Avoidance

“Which classmates would you not like to sit by?” ← Nominate up to 5

Binary directed network $y^{(k)}$, where $y_{ij}^{(k)} = \begin{cases} 1 & \text{if student } i \text{ avoids classmate } j \\ 0 & \text{otherwise} \end{cases}$

Friendship

(1) “Which classmates are your best friends?” ← Nominate up to 5

(2) “Which classmates do you often spend time with outside of school?”

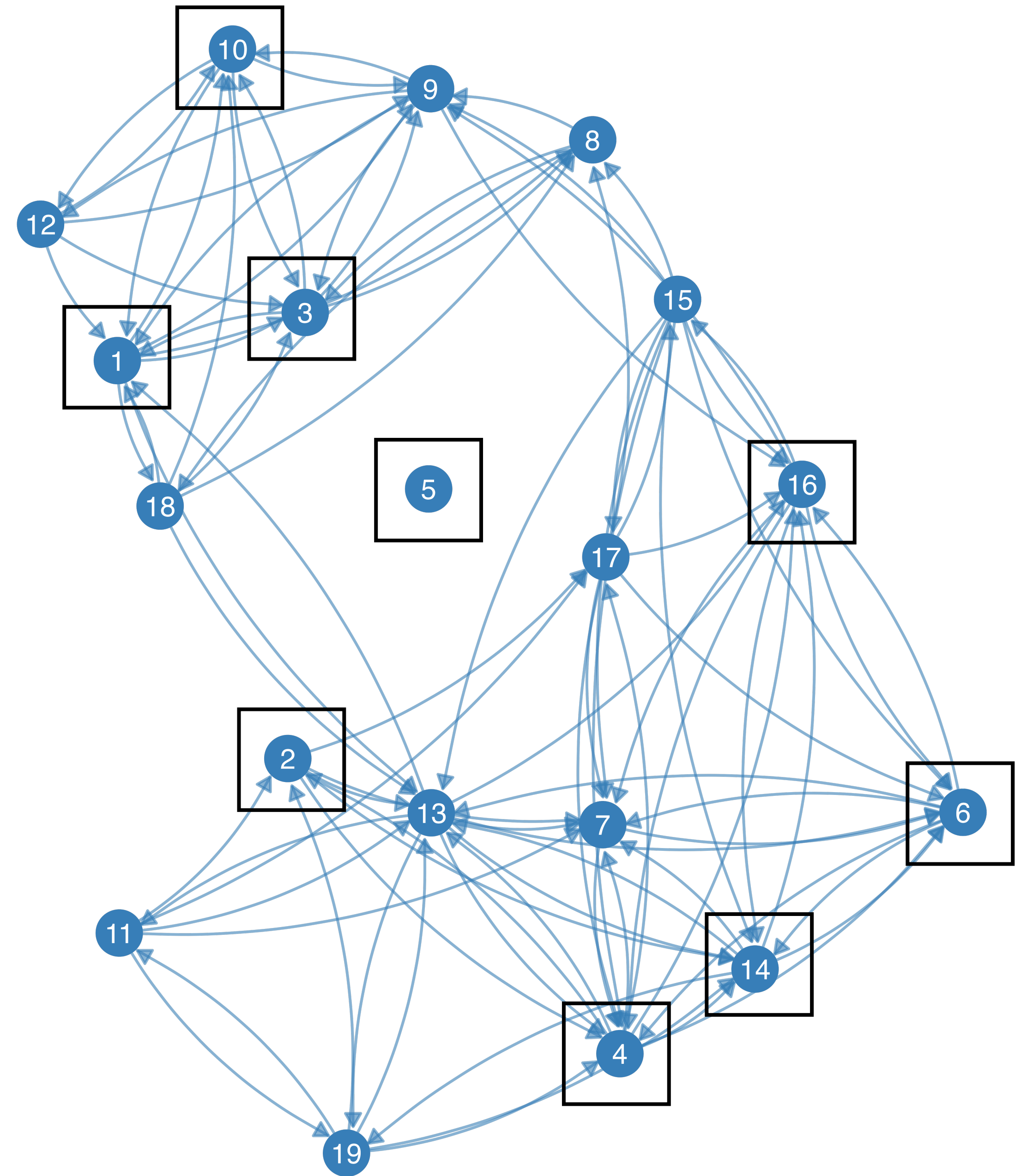
Student i nominates classmate j as a friend if they nominate them for (1) or (2)

Some students did not complete all the self-reported social relationship questions

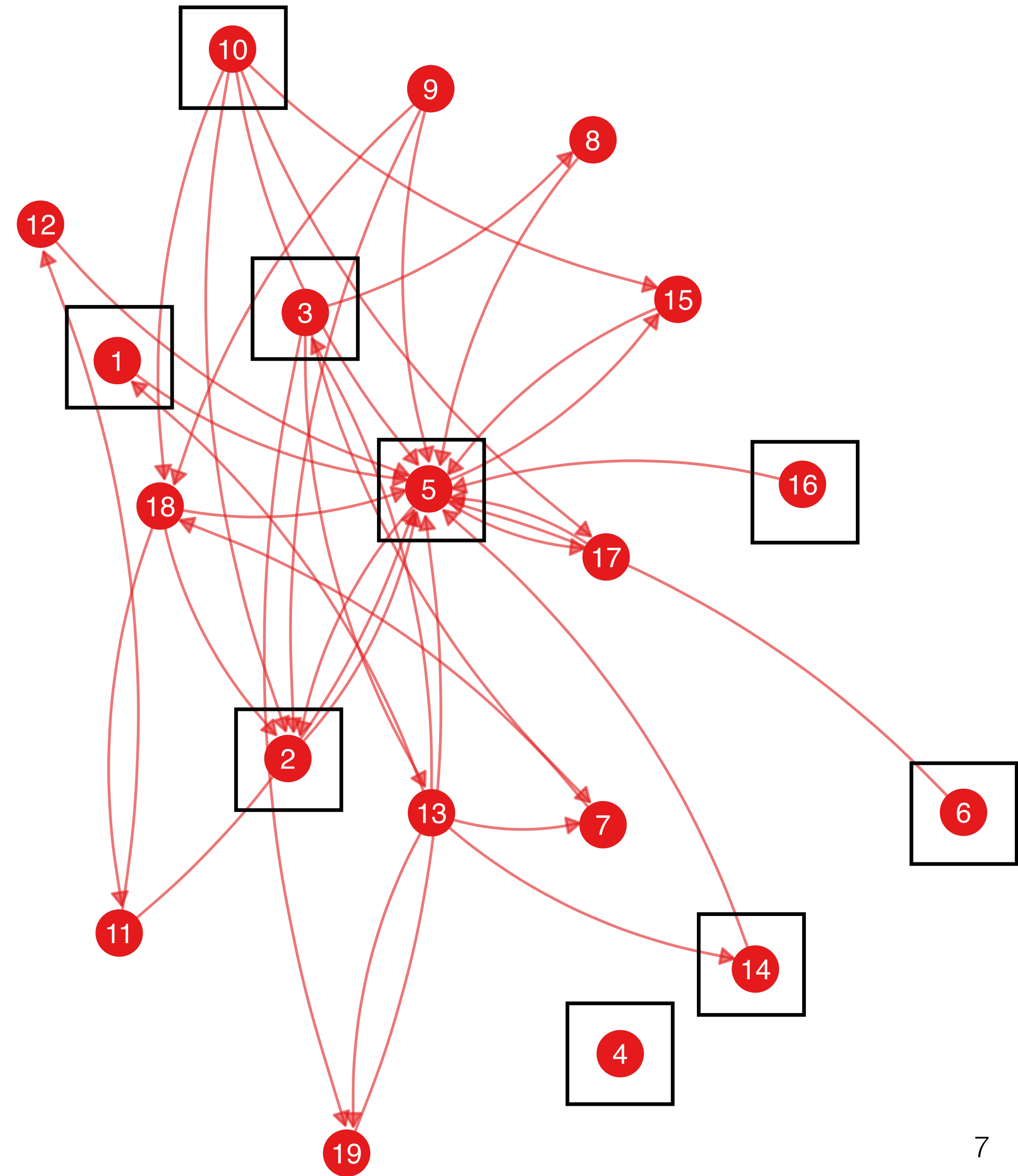
Question Type	Students Missing (%)
Not want to sit by	208 (5.1%)
Top 5 best friends	135 (3.3%)
Often spend time with outside of school	134 (3.3%)

We treat students who did not complete a given set of questions as having no nominations for that social relationship category

Friendship

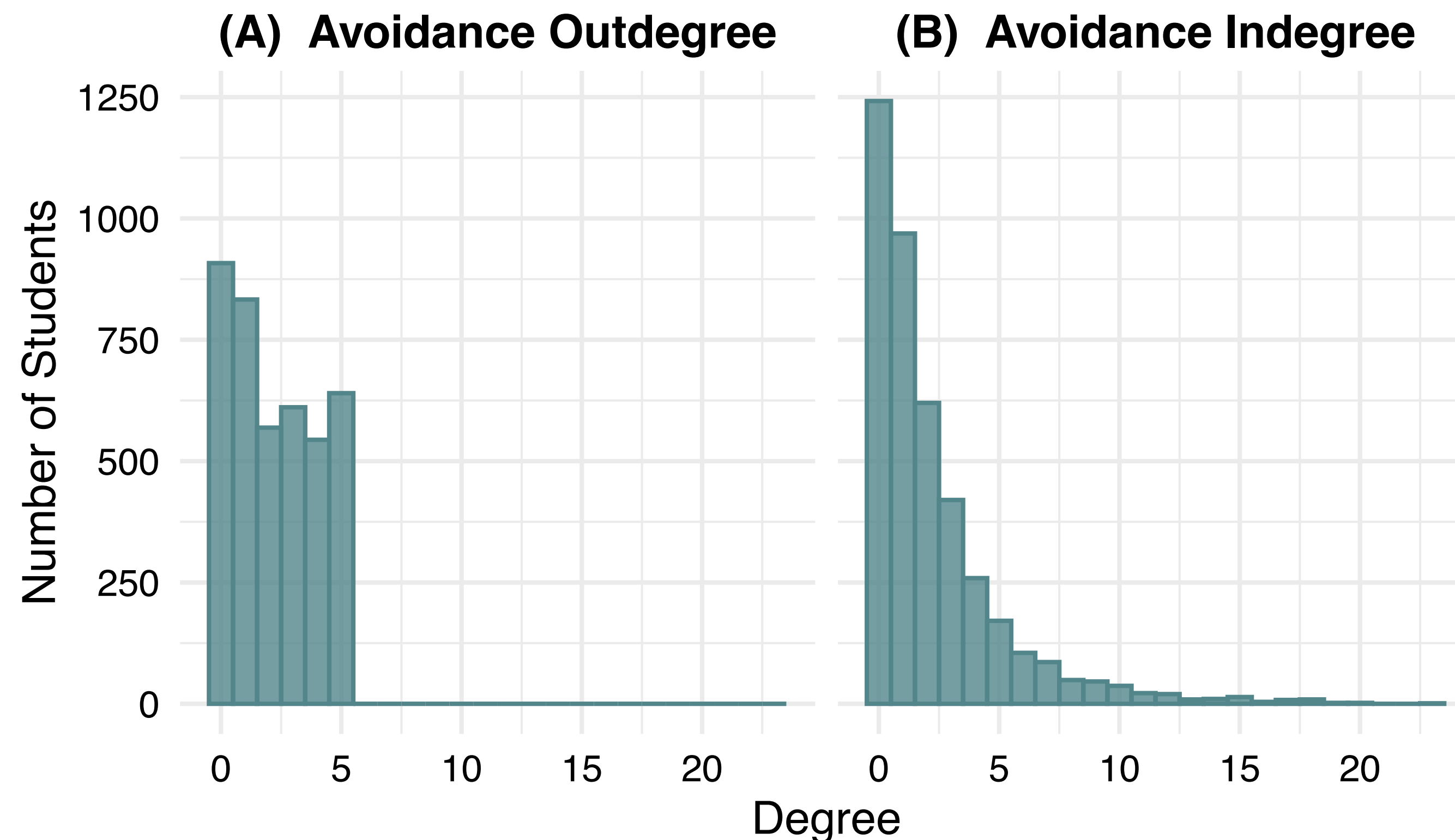


Avoidance



Degree-related descriptives

- Students avoid **2.4** ($SD = 1.7$) classmates and consider **3.7** ($SD = 1.7$) classmates their friends, on average
- **19.1%** of avoidance ties are reciprocated in a classroom, on average
- **7.4%** of students neither avoid nor are avoided by any of their classmates, on average



There is little overlap between friendship and avoidance

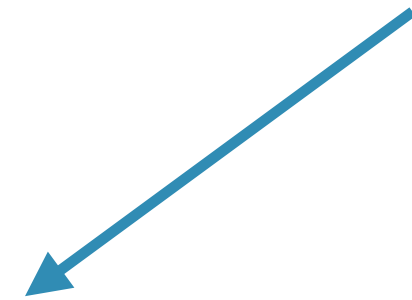
	Mean Jaccard Index	SD Jaccard Index
Avoidance and Friendship	0.023	0.029
Avoidance and Transposed Friendship	0.027	0.026

Transposed Friendship:

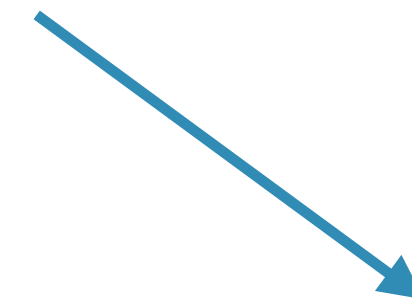
If classmate j considers student i a friend $\Rightarrow$ tie from student i to classmate j

Exponential Random Graph Models (ERGMs)

- Observed network is a single realization of a random network among a set of actors
- Observed network structure emerges from a series of local tendencies



Reciprocity: tendency for ties to be mutual between actors



Homophily: tendency for actors with the similar characteristics to form ties

- Outcome can be represented as the log-odds that actor i sends a tie to actor j , holding the other ties in the network fixed

Exponential Random Graph Models (ERGMs)

Observed network y is a realization of a random network $\mathbf{Y}$ with n actors

Conditional probability of observing y follows an exponential family distribution:

$$\mathbb{P}_{\theta, \mathcal{Y}}(\mathbf{Y} = y \mid \mathbf{X}) = \frac{\exp(\theta' g(y, \mathbf{X}))}{\kappa(\theta, \mathcal{Y})}$$

- $\mathbf{X}$: covariate matrix
- $g(y, \mathbf{X})$: sufficient statistic vector
- θ : coefficient vector for $g(y, \mathbf{X})$
- $\kappa(\theta, \mathcal{Y})$: normalizing constant computed over sample space of possible networks $\mathcal{Y}$

Contains counts of network configurations or a function of these counts within y and covariates in $\mathbf{X}$

e.g, reciprocity: $\frac{1}{2} \sum_i \sum_j y_{ij} y_{ji}$

Tie-level ERGM formulation

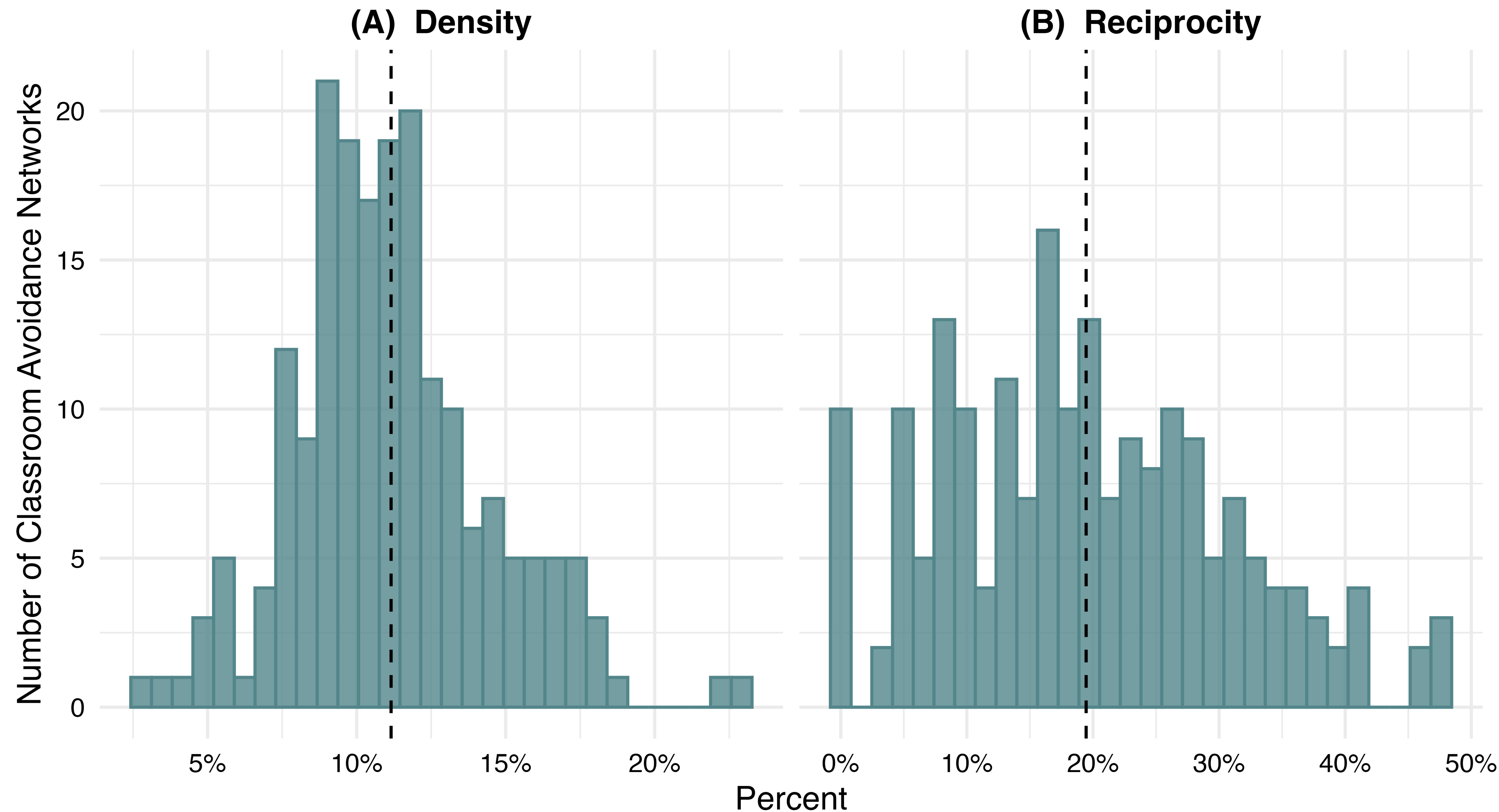
Outcome is the conditional log-odds of i sending a tie to j :

$$\log \left(\frac{\mathbb{P}(Y_{ij} = 1 \mid \mathbf{X}, Y_{-ij})}{\mathbb{P}(Y_{ij} = 0 \mid \mathbf{X}, Y_{-ij})} \right) = \theta' \delta_{ij}^+(y, \mathbf{X})$$

- Y_{ij} : tie from i to j
 - Y_{-ij} : all other ties in the network besides Y_{ij}
 - $\delta_{ij}^+(y, \mathbf{X})$: change statistic vector
- Measures change in sufficient statistic vector when $y_{ij} = 1$ vs. when $y_{ij} = 0$

ERGM coefficients: change in the conditional log-odds of a tie associated with a one-unit increase in the given statistic

Heterogeneity in classroom avoidance networks



Hierarchical ERGMs

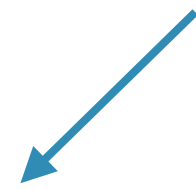
Extends the ERGM framework to a sample of networks:

- Allows the effects of the local tendencies on the network structure to vary across the sample
 - Results in network-specific random effects
- Assumes the network-specific random effects come from a common population distribution
 - “Borrows strength” across networks to estimate the network-specific random effects

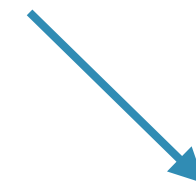
Hierarchical ERGM setup

Let $y^{(1)}, y^{(2)}, \dots, y^{(193)}$ denote our sample of classroom avoidance networks, where:

- $n^{(k)}$: number of students in the avoidance network for classroom k
- $\mathbf{X}^{(k)}$: exogenous covariates for classroom k



Student-level attributes
(e.g., student i 's gender)



Dyad-level attributes (e.g., whether
student i is friends with classmate j)

Hierarchical ERGM formulation

Classroom avoidance network $y^{(k)}$ is a realization of a random network $\mathbf{Y}^{(k)}$ with $n^{(k)}$ actors

Conditional probability of observing $y^{(k)}$ follows an exponential-family distribution:

$$\mathbb{P}_{\mathcal{Y}^{(k)}}(\mathbf{Y}^{(k)} = y^{(k)} \mid \mathbf{X}^{(k)}, \theta^{(k)}) = \frac{\exp\{(\theta^{(k)})^\top g(y^{(k)}, \mathbf{X}^{(k)})\}}{\kappa(\theta^{(k)}, \mathcal{Y}^{(k)})}$$

- $g(y^{(k)}, \mathbf{X}^{(k)})$: sufficient statistic vector for classroom k
- $\theta^{(k)}$: ERGM parameters specific to classroom k
- $\kappa(\theta^{(k)}, \mathcal{Y}^{(k)})$: normalizing constant for classroom k computed over sample space of possible networks $\mathcal{Y}^{(k)}$

Can contain counts of network configurations and exogenous covariates

e.g. reciprocity: $\frac{1}{2} \sum_i \sum_j y_{ij}^{(k)} y_{ji}^{(k)}$

Hierarchical ERGM formulation

While $\theta^{(k)}$ are specific to classroom k , we assume that for all $k \in \{1, \dots, 193\}$:

$$\theta^{(k)} \sim \text{Normal}(\mu, \Sigma) \quad \leftarrow \text{hierarchical part}$$

Thus,

- μ represents the average effect across our 193 classroom avoidance networks
- Main diagonal of Σ gives the variance of each effect across our 193 classroom avoidance networks

We focus most of our interpretations on μ and its variance (on main diagonal of Σ)

Model specification

In our hierarchical ERGM, we include:

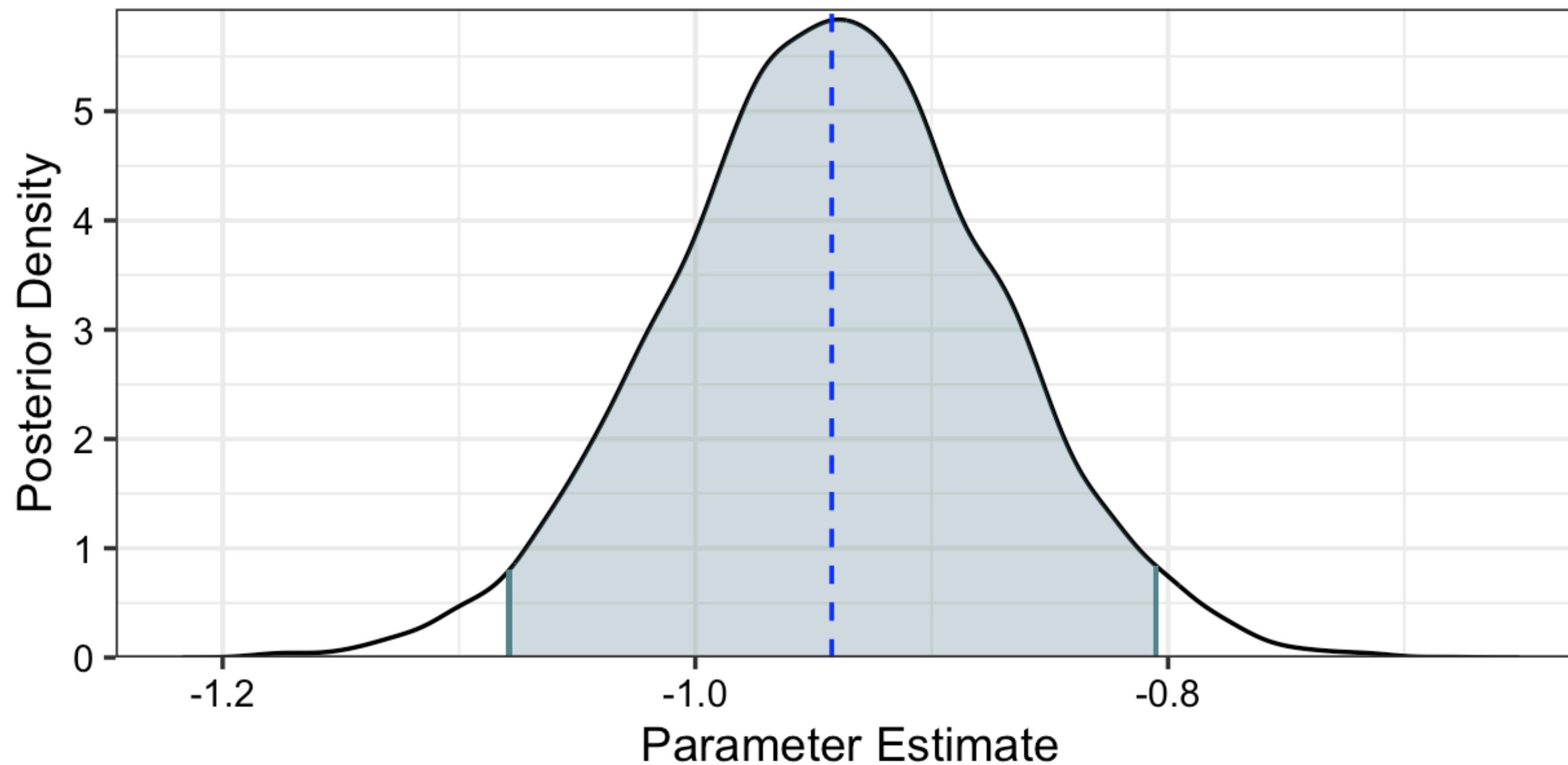
- (1) Friendship-related effects
- (2) Endogenous network effects (e.g., reciprocity, degree-related controls)
- (3) Gender controls (Kros et al., 2021)

Use Bayesian framework to estimate our parameters (Agneessens et al., 2024)*

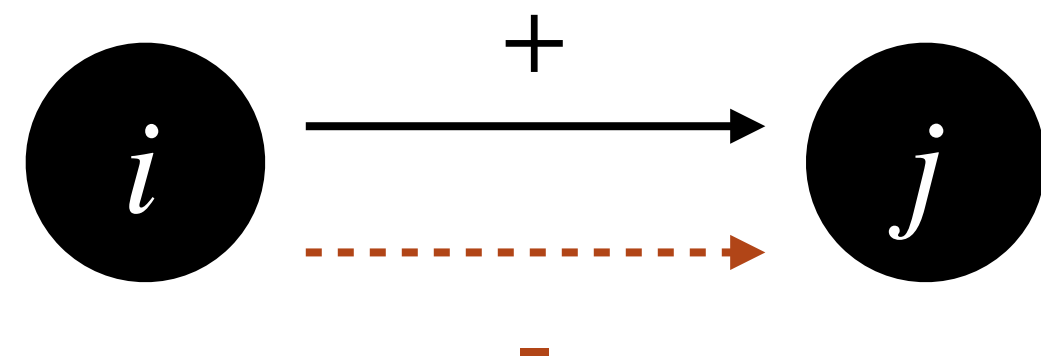
*<https://github.com/johankoskinen/BayesERGM>

Hierarchical ERGM estimation

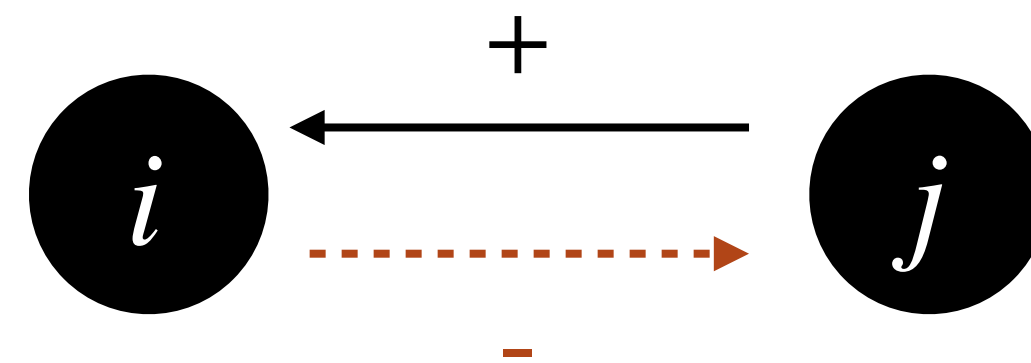
- (1) Obtain posterior distribution for each of our parameters
- (2) Extract each parameter's **posterior mean** and **95% credible interval (CI)**



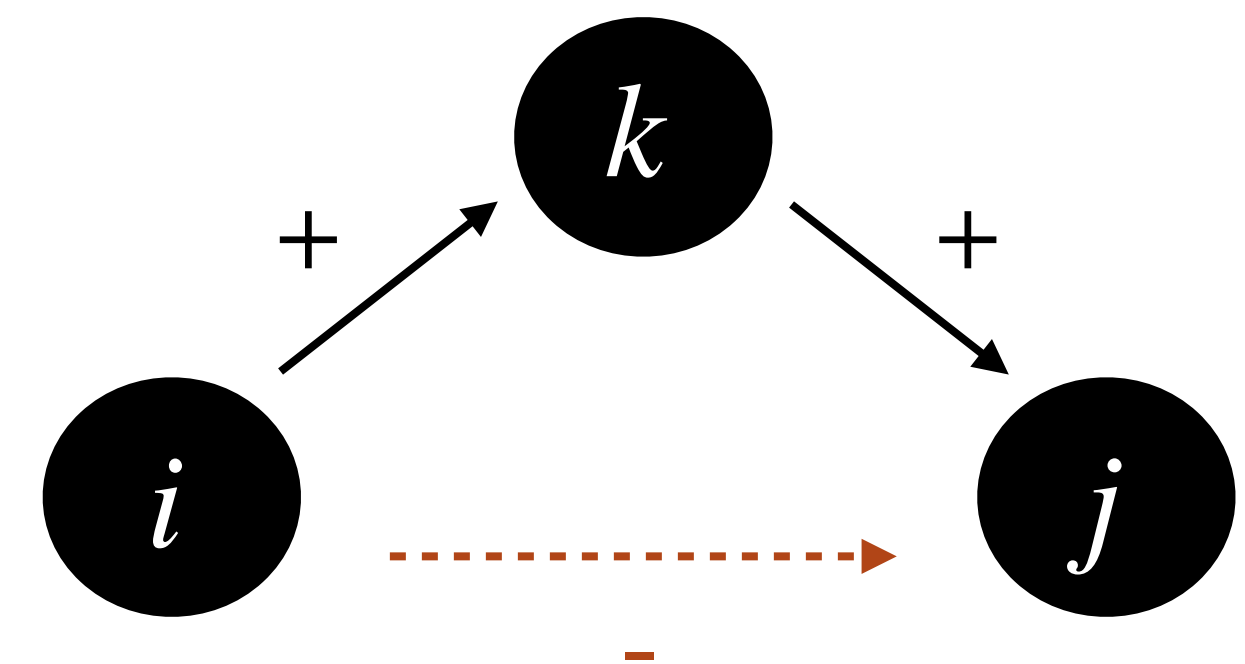
On average, avoidance networks are highly associated with friendship networks



Students tend not to avoid classmates they consider a friend (OR = 0.37, 95% CI [0.32, 0.44])

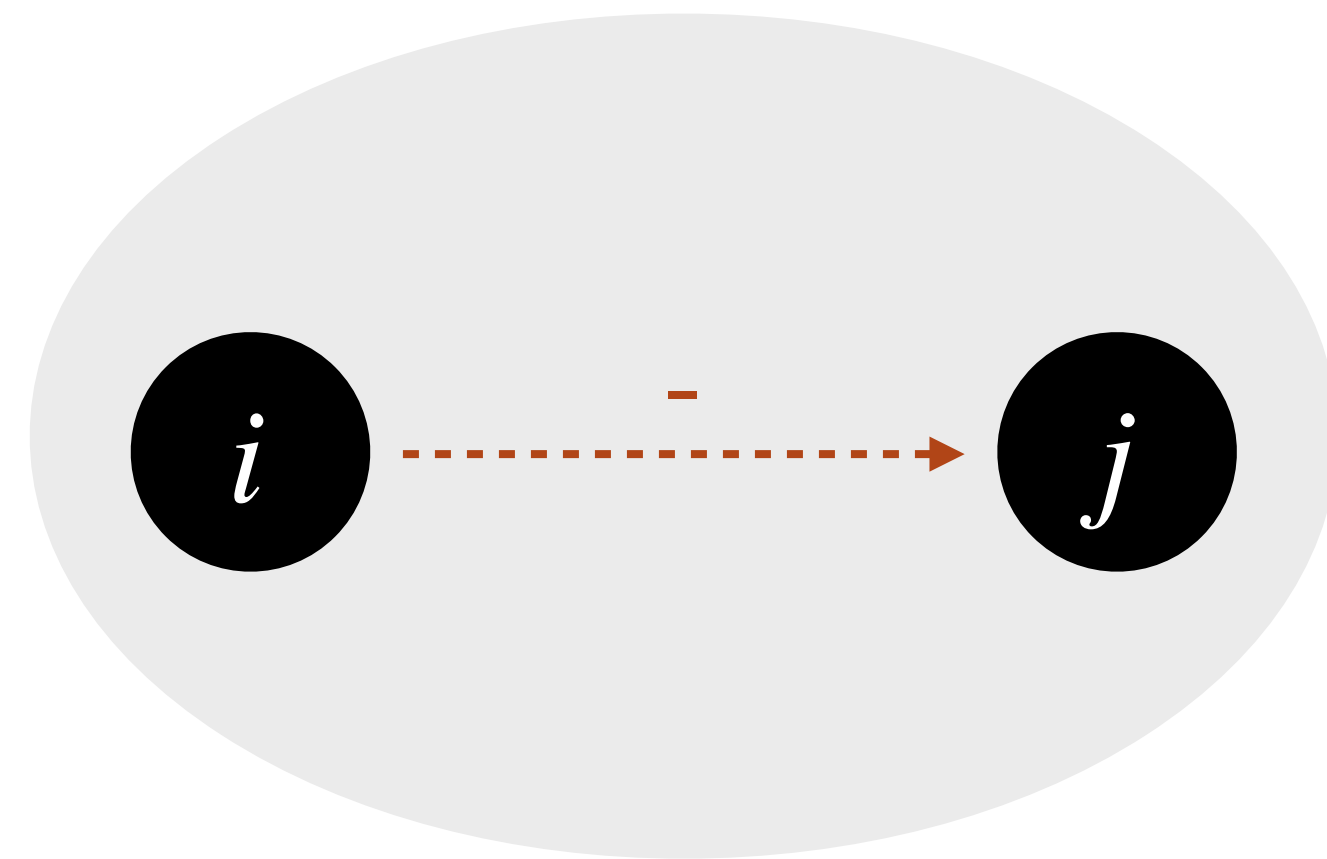


Students tend not to avoid classmates who consider them a friend (OR = 0.48, 95% CI [0.45, 0.53])

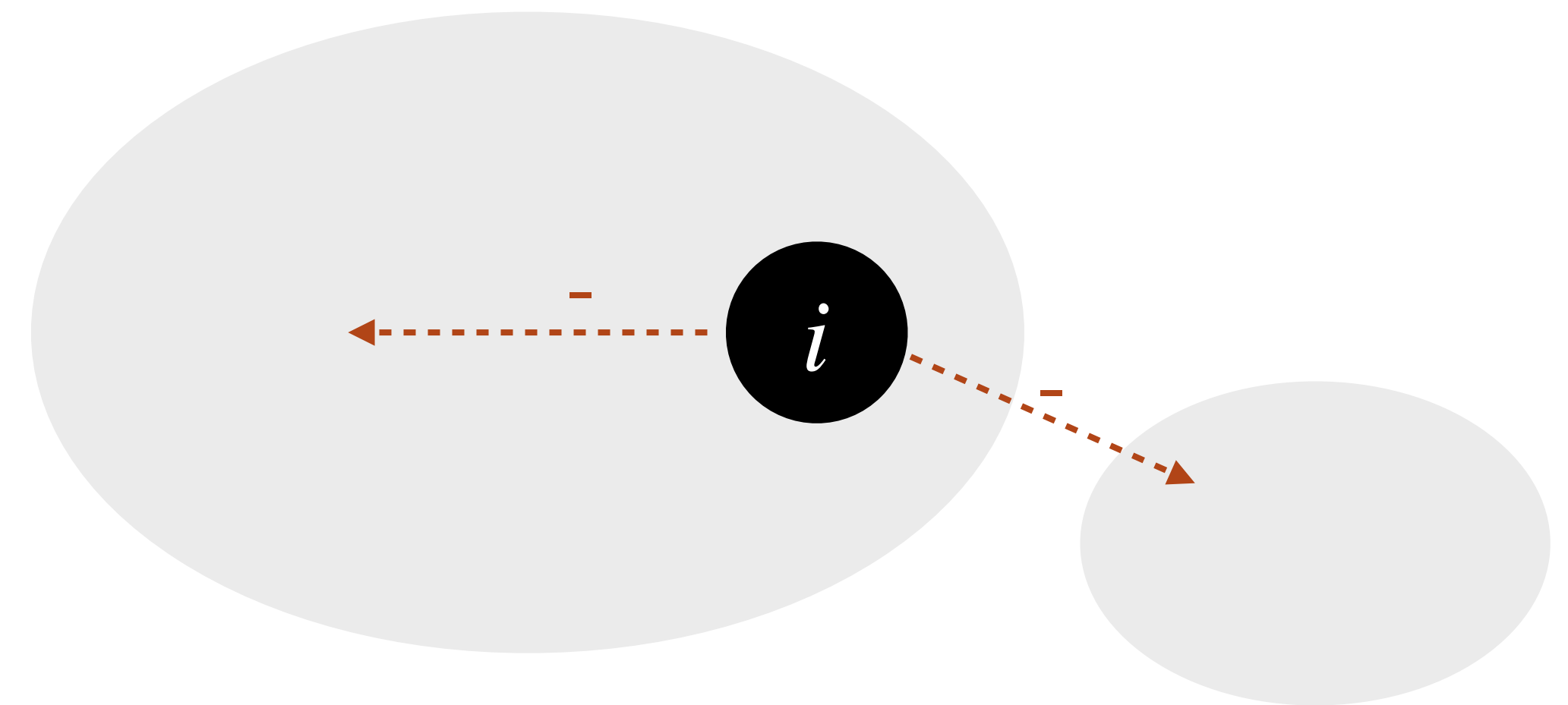


Students tend not to avoid classmates who are "friends-of-friends" (OR = 0.75, 95% CI [0.72, 0.79])

On average, avoidance networks are highly associated with friendship networks

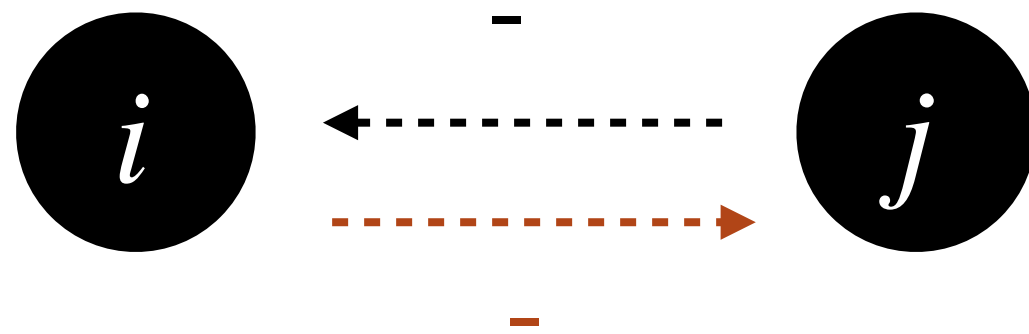


Students tend not to avoid classmates they are weakly connected to in the friendship network (OR = 0.69, 95% CI [0.63, 0.77])

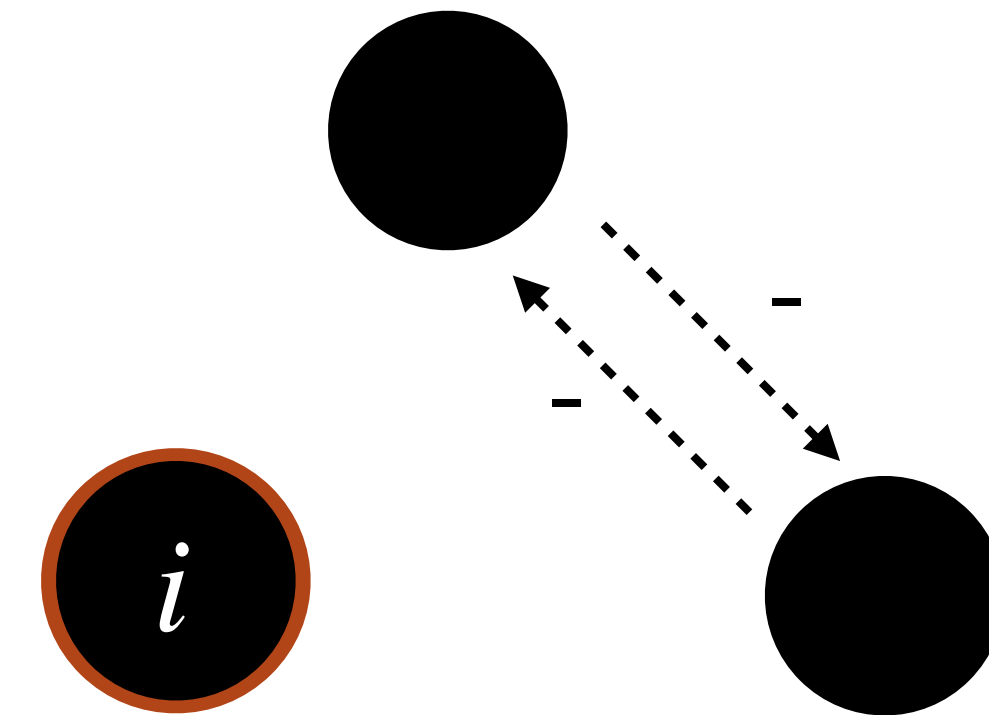


Students in the largest weakly connected component of the friendship network are more likely to avoid others (OR = 1.21, 95% CI [1.09, 1.33])

There are strong average reciprocity and degree-related endogenous effects in avoidance networks

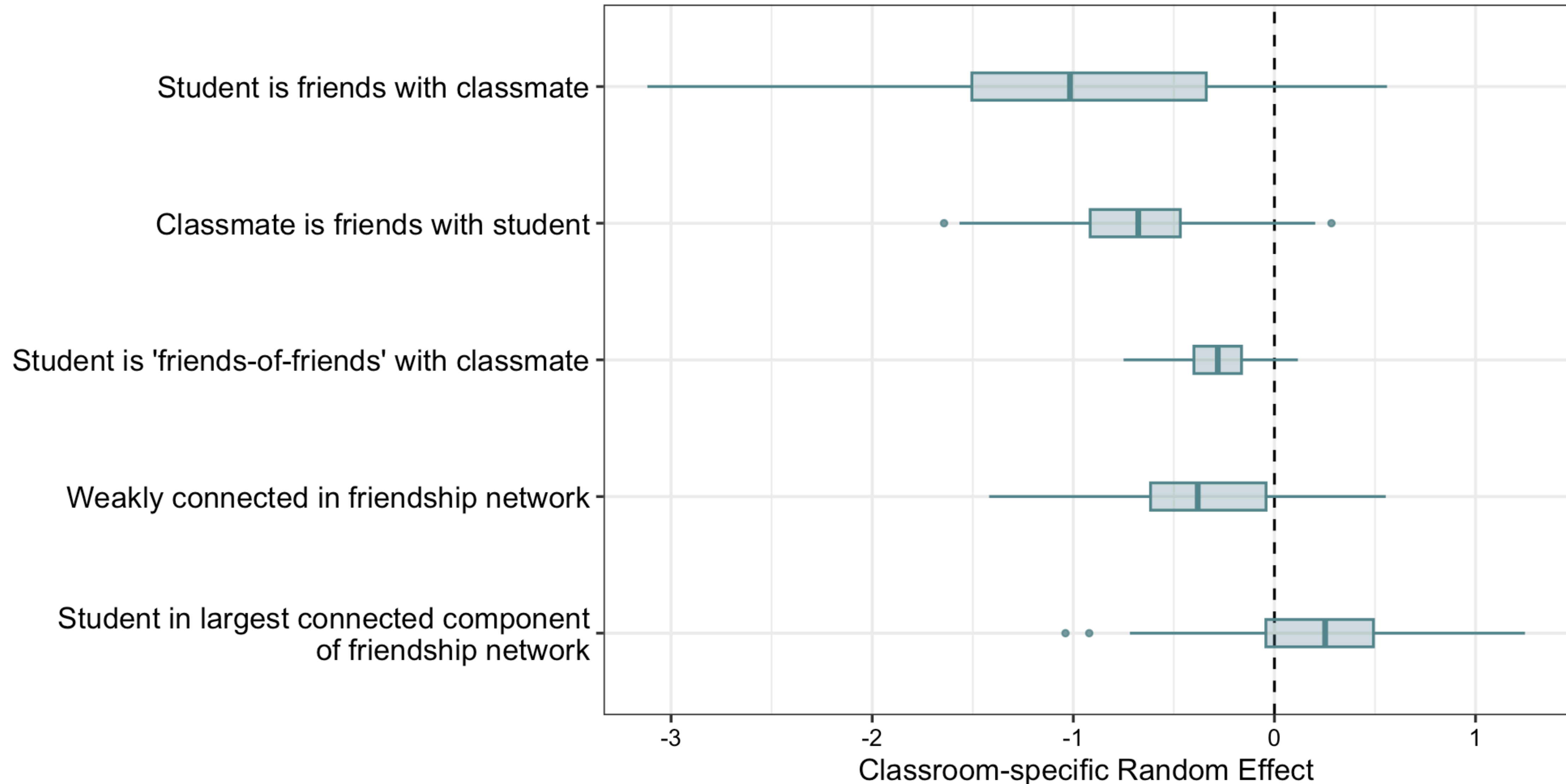


Avoidance tends to be reciprocated
(OR = 1.83, 95% CI [1.68, 1.99])



Many students avoid none of their
classmates (OR = 3.08, 95% CI [2.68, 3.54])

Variability in friendship effects across classrooms



Conclusions and future work

On average, classroom avoidance networks:

- (1) are highly associated with classroom friendship networks
- (2) exhibit strong reciprocity and degree-related endogenous effects

However, the magnitude of these effects can vary substantially between classrooms

E.g., we estimate that students in the largest connected component of the friendship network are less likely to avoid others in 54 (38%) of the classrooms

Future work: evaluating goodness-of-fit

Thank you! Questions?

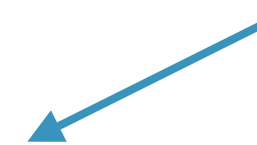
Appendix Slides

Bayesian estimation details

The Bayesian framework described by Agneessens et al. (2024) is a two-part Markov Chain Monte Carlo (MCMC) Algorithm

- (1) Update $(\theta^{(1)}, \dots, \theta^{(193)})$ via approximate exchange algorithm (Caimo and Friel, 2011)
- (2) Update (μ, Σ) via Gibbs sampling

For part (1): For each classroom $k \in \{1, \dots, 193\}$

- (1) Propose $\theta_*^{(k)}$ as update to the current parameter vector $\theta_0^{(k)}$ Set maximum outdegree to 5

- (2) Simulate a hypothetical classroom avoidance network $y_*^{(k)}$ using $\theta_*^{(k)}$
- (3) Compare sufficient statistics from $y_*^{(k)}$ with the sufficient statistics from $y^{(k)}$

We accept/reject the proposed parameter vector $\theta_*^{(k)}$ based on this comparison

Bayesian estimation details

For part (2): Place a Normal-Inverse Wishart prior on (μ, Σ) :

$$\mu \mid \Sigma \sim \text{Normal}(0, \Sigma)$$

$$\Sigma \sim \text{InvW}(\Lambda, \nu)$$

where $\nu = 1$ and $\Lambda = \text{diag}(0.1)$

Currently, we have:

- 50,000 MCMC iterations
- 40,000 iterations are discarded (burn-in) and thinning factor of one

Last 10,000 samples are used to estimate the hierarchical ERGM parameters